

ECE 212 – LAB 3

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Introduction:

ASK stage and inspirations:

- I had the idea to design a circuit that would reproduce the Mustang's turning signals. I chose this project because I really like mustangs and got really excited with the idea of designing something with it.
- I decided to try to use a toy model to wire LEDs inside the car to have them light up the turning signals.

Research done:

- For this project I used a 555 timer to generate a clock signal. I used a circuit found in "Learning About Electronics" (<https://www.learningaboutelectronics.com/Articles/555-timer-clock-circuit.php>)

Planing:

Planing documents (notebook):

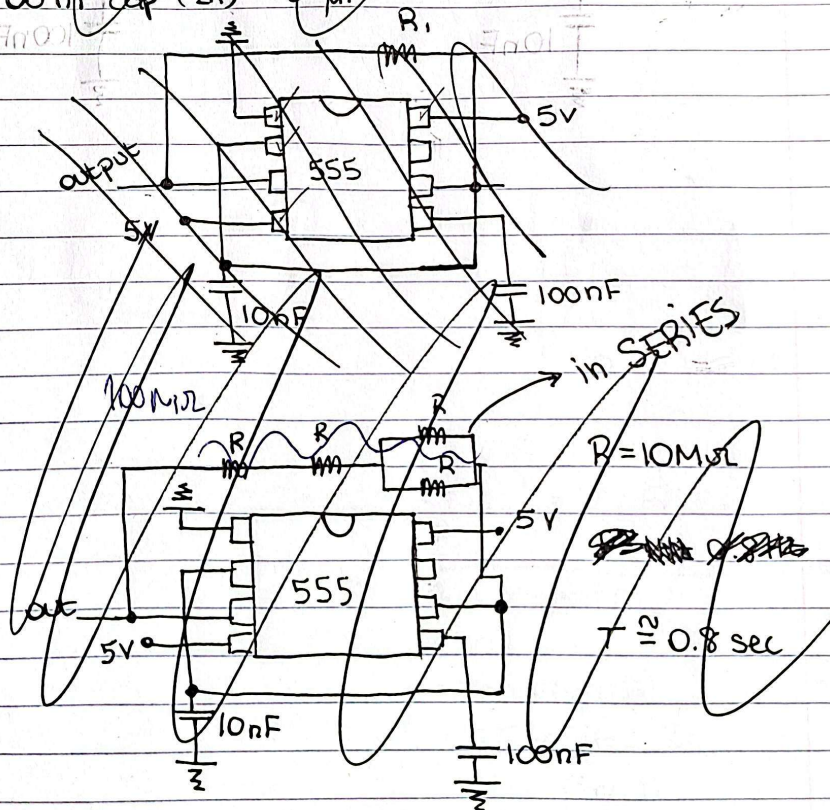
Clock Signal (using 555 timer)

"Learning about electronics" (source)
might need two clocks

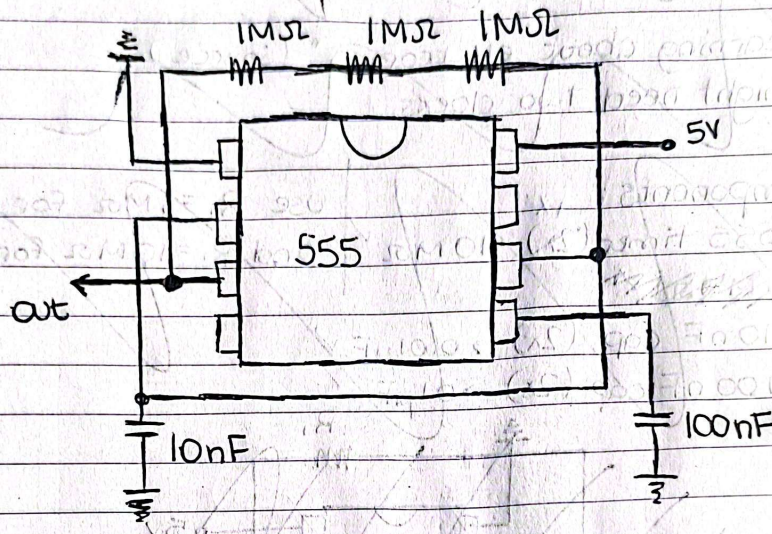
Components:

- 555 timer (2x)
- 10 M Ω
- 10 nF cap (2x) $\rightarrow$ 0.01 μ F
- 100 nF cap (2x) $\rightarrow$ 0.1 μ F

Use $R_1 = 1 \text{ M}\Omega$ for 60 Hz
and $R_1 = 10 \text{ M}\Omega$ for 6 Hz



Clock Signal (Using 555 timer)



Sequence 1: Turn Signal Right

LED1 LED2 LED3

4 states:

0	0	0
1	0	0
1	1	0
1	1	1

} 2 bit input

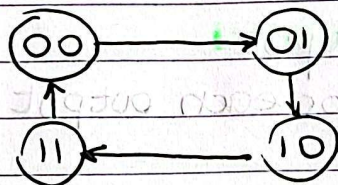
4 states



→ 2 D-Flip Flops (Sync Binary Counter)

Sync. Binary Counter:

State diagram:



current		next	
I_1	I_2	I_1^*	I_2^*
0	0	0	1
0	1	1	0
1	0	1	1
1	1	0	0

I_1^* :

$I_2 \backslash I_1$	0	1
0	0	1
1	1	0

$$I_1^* = I_2' I_1 + I_2 I_1' = I_1 \oplus I_2$$

I_2^* :

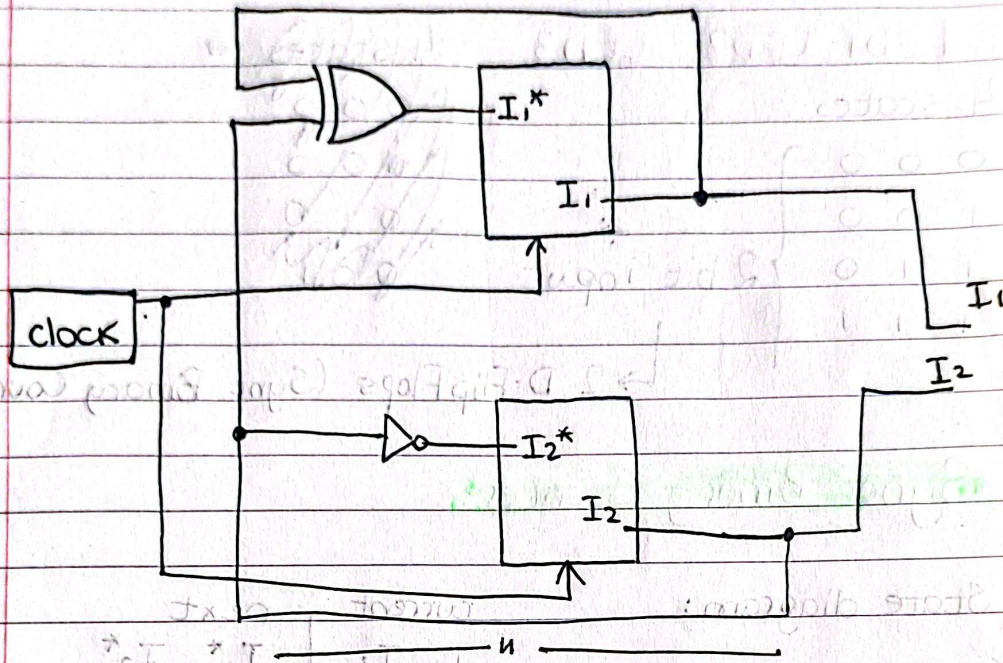
$I_2 \backslash I_1$	0	1
0	1	1
1	0	0

$$I_2^* = I_2'$$

Components:

- XOR
- INVERTER
- D Flip Flop 2x
- Clock Circuit

Circuit Diagram:

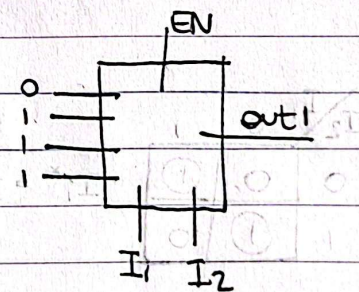


2 Bit input to 3 Bit output

will use 3 4:1 MUX $\rightarrow$ one for each output Bit

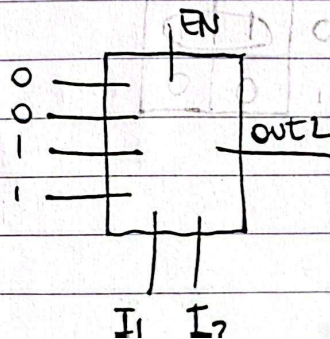
MUX 1:

I_1	I_2	Out 1
0	0	0
0	1	1
1	0	1
1	1	1



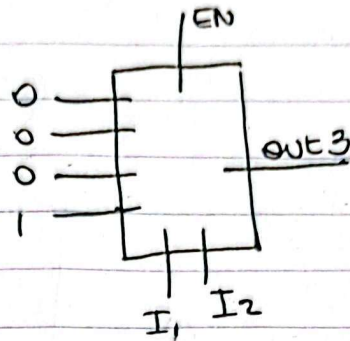
MUX 2:

I_1	I_2	Out 2
0	0	0
0	1	0
1	0	1
1	1	1

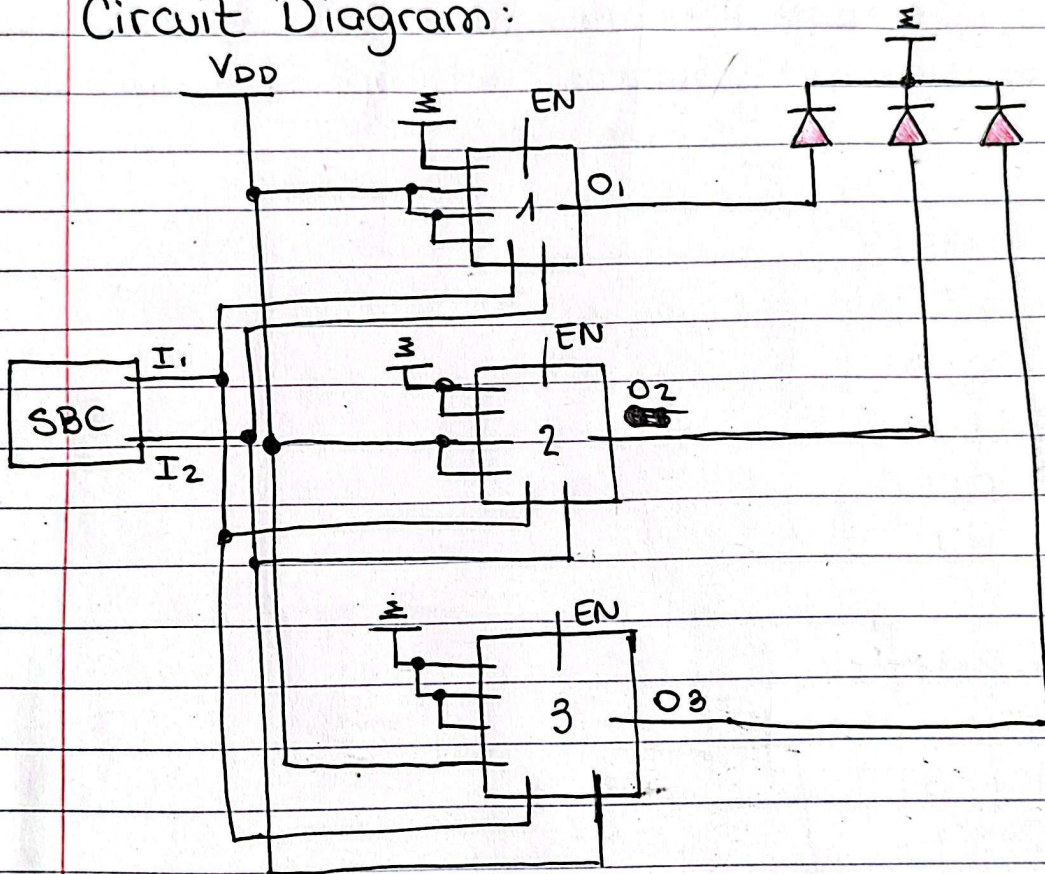


MUX 3:

I_1	I_2	Out 3
0	0	0
0	1	0
1	0	0
1	1	1



Circuit Diagram:



Components used:

- 4:1 MUX (3x)
- SBC circuit output
- Red LEDs (3x)

Sequence 2: Turn Signal - Left

Out3 Out2 Out1
LED3 LED2 LED1

4 states

000
001
011
111

using Sync Binary Counter already built
(run through the 4 states → 00, 01, 10, 11)

- will need to convert two input to three out.

- use 4:1 MUX.

LED 3:

	I ₁ , I ₂	Out3:
D ₀	0 0	0
D ₁	0 1	0
D ₂	1 0	0
D ₃	1 1	1

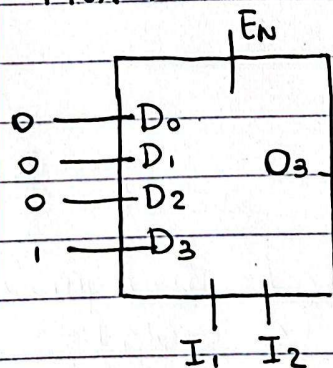
LED 2:

	I ₁ , I ₂	Out2:
D ₀	0 0	0
D ₁	0 1	0
D ₂	1 0	1
D ₃	1 1	1

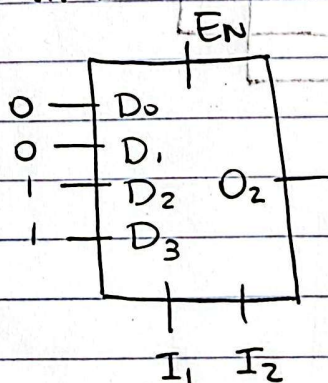
LED 1:

	I ₁ , I ₂	Out1:
D ₀	0 0	0
D ₁	0 1	1
D ₂	1 0	1
D ₃	1 1	1

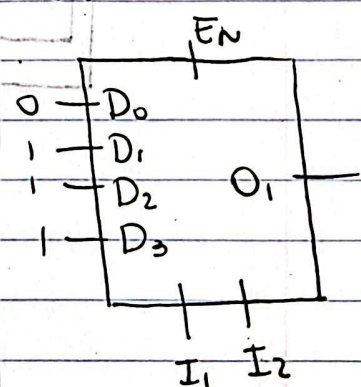
MUX 1:



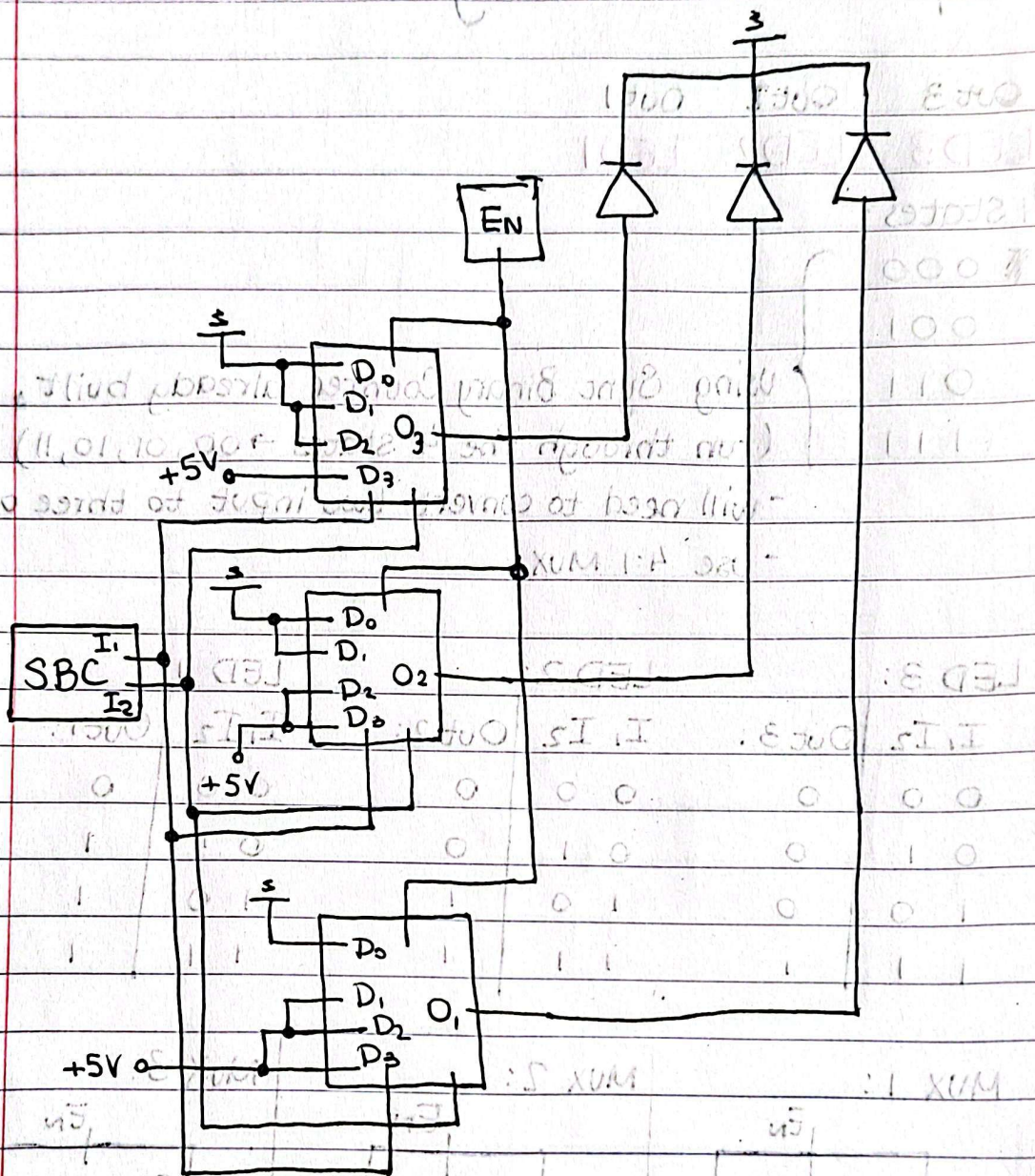
MUX 2:



MUX 3:



Sequence 2: Turn Signal - Left



Explanations:

- The first two pages show the planning done when I was trying to build a clock signal using a 555 timer. The different circuit diagram variations were different attempts to change the frequency of the output signal.
- In **Sequence 1: Right Turn Signal**, I have the planning on how I would have four states and convert those into a 3 bit output (one for each LED).
 - I decided to use a Synchronous Binary Counter to run through a “loop” of the four states. For the SBC I used two D Flip-Flops and an XOR gate. The planing and circuit diagram are shown in the previous section (document/notebook).
 - To convert the SBC circuit 2 bit output into a 3 bit output, I used some 4:1 MUX (one for each new output bit). Each one of the MUXes gave the output for a specific LED, where it would output a specific bit depending on the state received by the SBC.
- In **Sequence 2: Left Turn Signal**, I have the planning on how I would have four states and convert those into a 3 bit output (one for each LED).
 - I used the same design from Sequence 1 for the SBC, but then had a different circuit for my conversion from 2 bit to 3 bit (4:1 MUXes).

Parts used:

Clock signal:

Component	Uses
555 Timer	Base for the Clock Signal
10nF capacitor	
100nF capacitor	
3x 1M Ohm resistor	

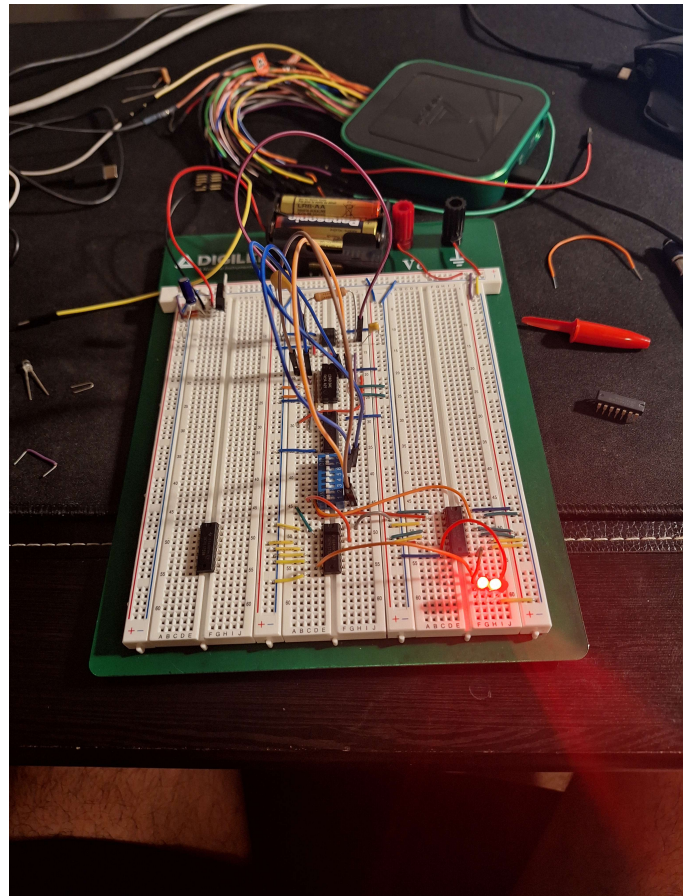
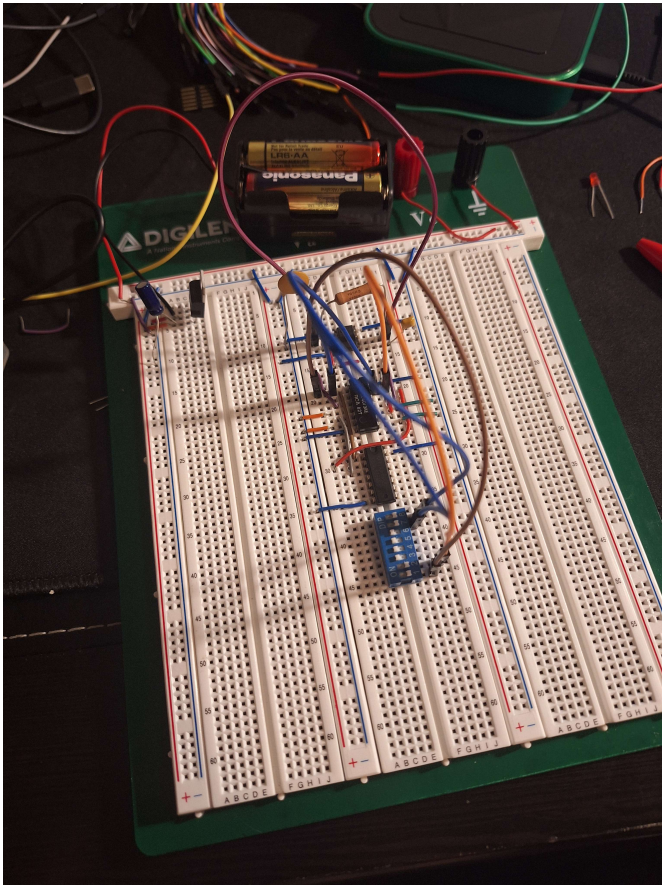
Synchronous Binary Counter:

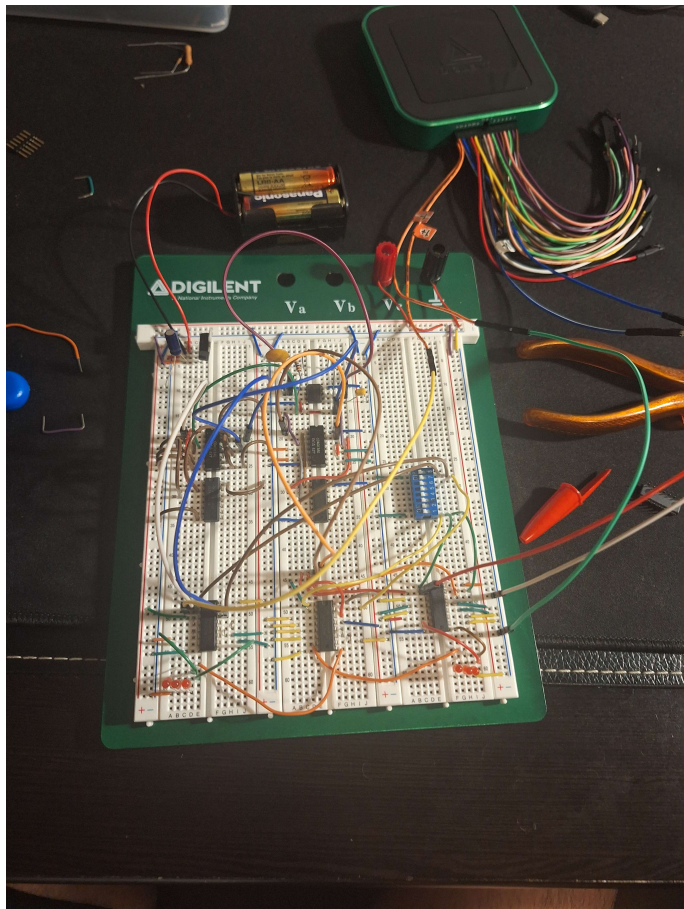
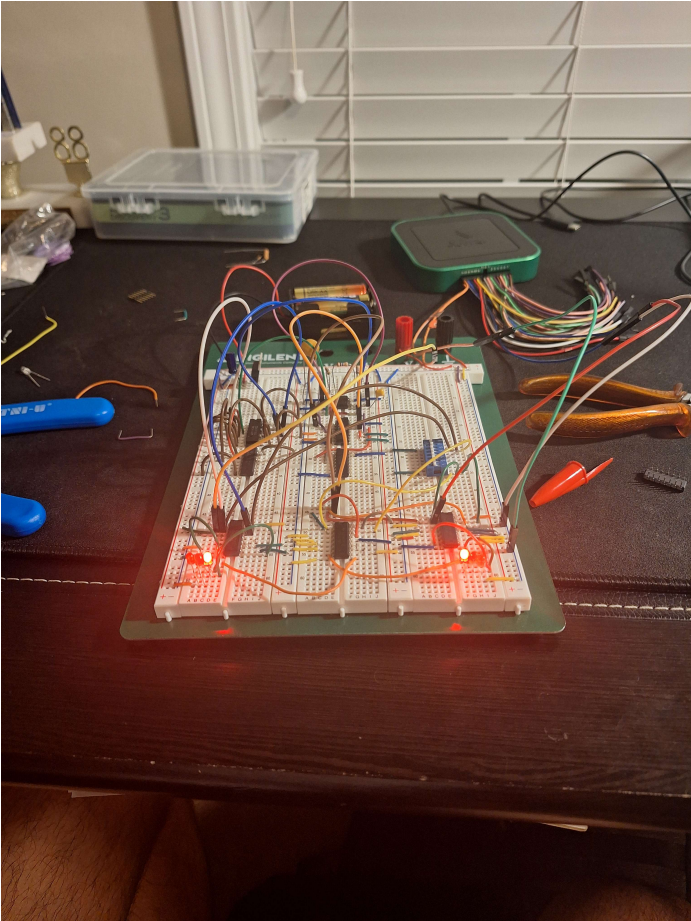
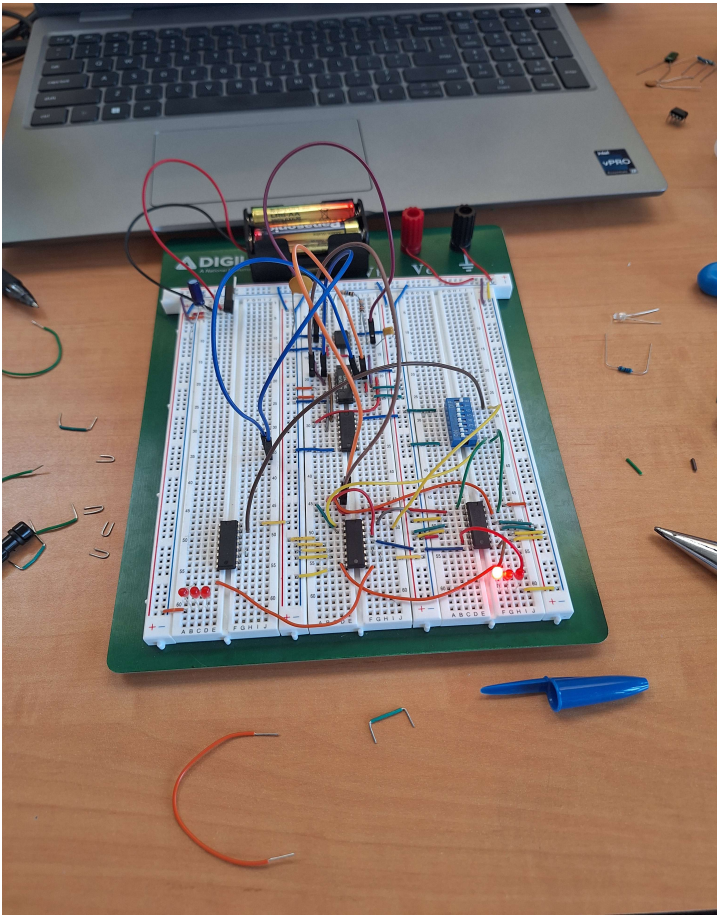
Component	Uses
D Flip-Flop (CD4013AE)	Responsible for changing the output according to the clock signal (each one gives us a bit of the state)
XOR Gate (SN74LS86AN)	Glue logic
Clock Signal	Using as input for the CLK in the Flip-Flop

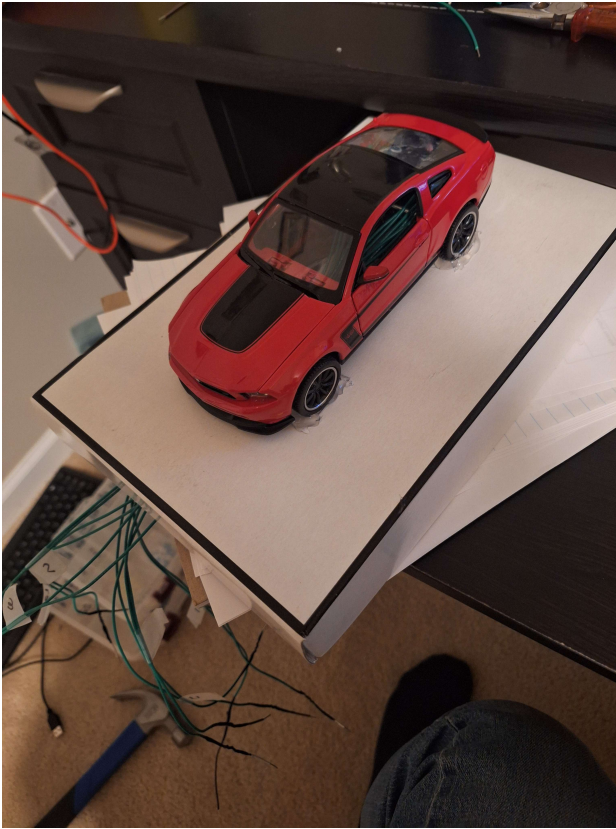
2 bit to 3 bit:

Component	Uses
3x 4:1 MUX (SN74LS86AN)	These are used to get the SBC state signal and convert the two bits into a single bit output that will be determining if the LED is on or off.
3x LED's	Used to display output
SBC signal	Used as select bits in the MUXes

Photos and Videos of Progress:







Demonstration Video (only one turn signal):

<https://drive.google.com/file/d/1mpwDF4IjMZmKDKGqtGofGTCKqJBErNzH/view?usp=sharing>

(The third LED is dying but it still lights up)

Changes and Improvements:

- I had to iterate and change the circuit for the clock signal many different times as I wasn't getting the period I wanted for this project.
- I planned and designed one side of the circuit (Right turn signal) and then I built that circuit to make sure that it was working, then I planned and designed the left side of the circuit.
- To finalize my project I soldered the LEDs to long jump wires, and ran them through the model car. I then glued the car on top of a cardboard box and put my breadboard inside the box, where the wires will be connected.
- To better improve my design I will solder jump wires to my DIP switch (controls what turning signal to turn on) and glue the switch to the box cover next to the car.

Final Demonstration:

https://drive.google.com/file/d/1KQ6-OgdYB-7VI4nv1mfmXan2siB_n-C_/view?usp=sharing

